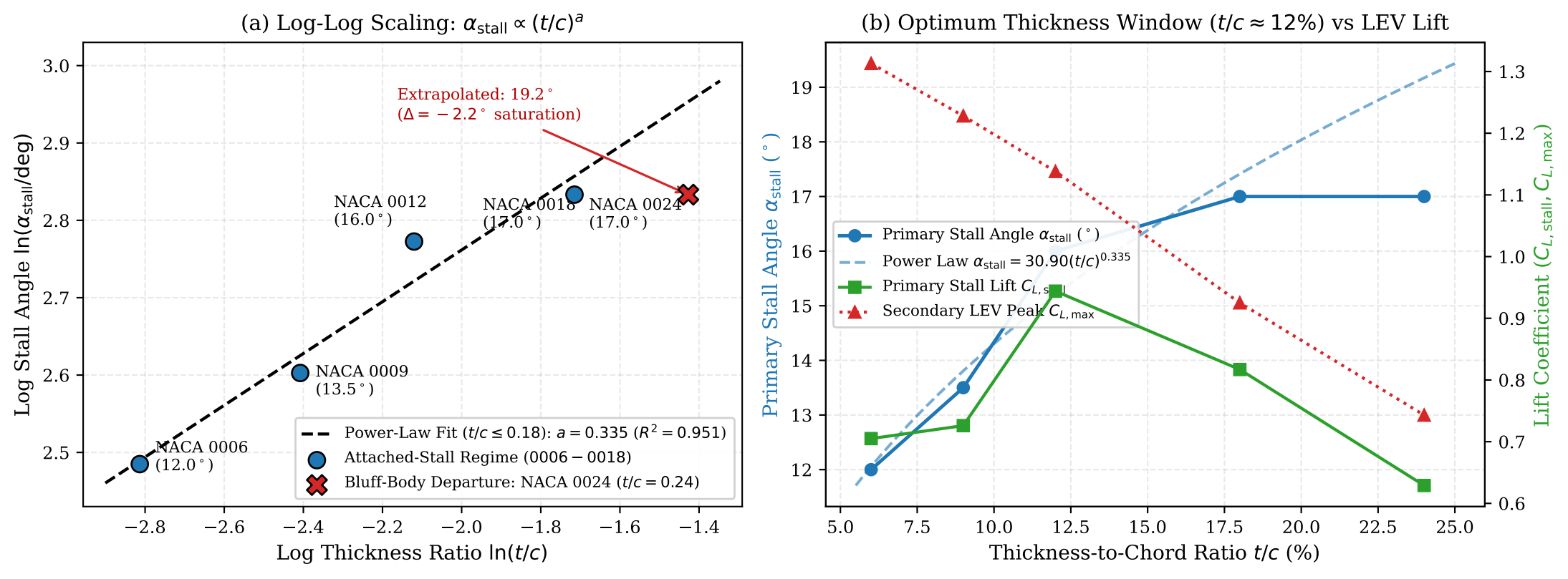
(a) Log-Log Scaling: $\alpha_{stall} \propto (t/c)^a$
Log Stall Angle $\ln(\alpha_{stall}/\mathrm{deg})$
Log Thickness Ratio $\ln(t/c)$
Extrapolated: 19.2°
(Δ = −2.2° saturation)
NACA 0012
(16.0°)
NACA 0018
(17.0°)
NACA 0024
(17.0°)
NACA 0009
(13.5°)
NACA 0006
(12.0°)
Power-Law Fit ($t/c \le 0.18$): $a = 0.335$ ($R^2 = 0.951$)
Attached-Stall Regime (0006 − 0018)
Bluff-Body Departure: NACA 0024 ($t/c = 0.24$)
(b) Optimum Thickness Window ($t/c \approx 12\%$) vs LEV Lift
Primary Stall Angle α_{stall} (°)
Power Law $\alpha_{stall} = 30.90(t/c)^{0.335}$
Primary Stall Lift $C_{L,stall}$
Secondary LEV Peak $C_{L,max}$
Primary Stall Angle α_{stall} (°)
Lift Coefficient ($C_{L,stall}$, $C_{L,max}$)
Thickness-to-Chord Ratio t/c (%)